

Sprint 2 Review and Retrospective – Ctrl + Incision

1. What went well

Went Well Item 1: End-to-End Tracked Ultrasound Pipeline Integration

Description (what worked and why):

The team successfully integrated live ultrasound acquisition and 6-DOF tracking using PlusServer, producing synchronized tracked ultrasound .mha sequences. These sequences were validated through offline volume reconstruction using PlusToolkit's Volume Reconstructor and visualized using VTK. The modular separation between acquisition, reconstruction, and visualization allowed each stage to be tested independently, improving debugging efficiency and system stability.

Went Well Item 2: 3D Volume Reconstruction and Improved Visualization Interface

Description (what worked and why):

The offline 3D reconstruction workflow was successfully implemented using PlusToolkit and SlicerIGT, enabling conversion of tracked ultrasound sequences into volumetric datasets. Reconstruction was driven through scripted pipelines (Python subprocess and batch execution), allowing consistent and repeatable model generation. The visualization interface was improved to better support switching between 2D tracked ultrasound frames and reconstructed 3D volumes, making spatial consistency and reconstruction quality easier to inspect. These UI improvements enhanced usability and supported clearer validation of reconstruction results.

Went Well Item 3: ArUco-Based Stylus Tracking Integration

Description (what worked and why):

ArUco-based stylus tracking was successfully implemented using calibrated camera intrinsics. The system detects ArUco markers, estimates full 6-DOF pose, and overlays visual indicators (marker outlines, axes, and center point) directly onto the live camera feed. This provided immediate visual validation of pose estimation correctness and ensured the stylus tracking subsystem could be tested independently from the ultrasound and tracking hardware pipeline.

Went Well Item 4: Separation of Online and Offline Calibration Workflows

Description (what worked and why):

Temporal calibration and volume reconstruction were performed offline using PlusToolkit tools, while camera-based stylus tracking was handled in real time within the application. This

separation reduced coupling between subsystems and allowed each calibration workflow (camera intrinsic, temporal synchronization, stylus tracking) to be validated using appropriate tools without blocking overall system progress.

Went Well Item 5: Clear Task Decomposition Across Calibration and Reconstruction Pipeline

Description (what worked and why):

The team effectively decomposed the complex calibration and reconstruction workflow into discrete, well-defined tasks. Ultrasound video preparation, temporal alignment, data formatting, reconstruction execution, and visualization were handled independently, reducing cross-dependencies and enabling parallel development. This structure helped isolate issues more efficiently and ensured steady progress across the sprint.

2. To Improve

To Improve Item 1: Calibration Workflow Complexity and Tooling Overhead

Description (what did not work / pain point):

Configuring and running PlusToolkit calibration pipelines required significant manual setup, including XML configuration management, correct executable selection, and careful data formatting. Small configuration mismatches (paths, orientation flags, executable usage) often resulted in silent failures or unclear error messages, increasing debugging time.

To Improve Item 2: Limited Automation of Stylus Calibration (Axial & Pivot)

Description (what did not work / pain point):

While real-time ArUco pose tracking was implemented and validated, full axial and pivot calibration solvers were not fully automated within Sprint 2. Calibration data collection and mathematical solving steps remain partially manual and require further integration to produce finalized calibration transforms usable by downstream systems

To Improve Item 3: Verification and Debugging of 3D Reconstruction

Description (what did not work / pain point):

Although 3D volume reconstruction was successfully performed, verifying reconstruction correctness required manual inspection across multiple tools. There was limited automated validation to confirm spatial alignment, correct temporal synchronization, and expected volume bounds. As a result, identifying whether reconstruction issues originated from tracking data, calibration, or reconstruction parameters was time-consuming.

3. Not Done Items from Sprint 2

Not Done Item 1: Automated Axial and Pivot Stylus Calibration

Reason not completed:

Scope prioritization and dependency on stable real-time pose estimation limited the ability to fully implement and validate automated axial and pivot calibration solvers within the sprint timeframe.

Plan:

Extend the ArUco tracking module to log pose samples and implement least-squares pivot and axial calibration solvers prior to pre-design review. These solvers will generate reusable stylus tip, and axis transforms compatible with PlusToolkit and VTK visualization.

Not Done Item 2: Automated Validation of 3D Reconstruction Accuracy

Reason not completed:

Sprint 2 focused on enabling successful volume reconstruction and visualization rather than quantitative validation. While reconstructed volumes could be visually inspected in VTK and Slicer, no automated metrics (e.g., spatial error, coverage completeness, or consistency checks) were implemented to formally assess reconstruction quality.

Plan:

Before pre-design review, implement basic validation checks such as volume bounds verification, frame coverage statistics, and optional landmark-based alignment checks using known phantom geometry.